

An Exploratory Comparative Analysis of Poetic Phonological Density

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Abstract—This study presents an exploratory computational analysis of phonological characteristics in poetic texts. Using syllable and stress-based measures derived from the CMU Pronouncing Dictionary, the analysis examines patterns of phonological density and rhythmic structure across multiple poetic corpora. The findings reveal clear differences in phonological organisation and rhythmic regularity, demonstrating the effectiveness of computational phonological methods for the quantitative study and comparison of poetic style. The present study was conducted as the first assignment for the *Data Analysis for Humanities with Python* course of the MSc Digital Humanities programme, jointly offered by the *National and Kapodistrian University of Athens*, the *University of Cyprus*, and the *ATHENA Research Centre*.

Index Terms—Digital Humanities, Computational Linguistics, Phonological Density, Natural Language Processing, CMU Pronouncing Dictionary

I. INTRODUCTION

The present study conducts an exploratory computational analysis of the phonological characteristics of selected poems by William Shakespeare, Emily Dickinson, and Elizabeth Barrett Browning. The primary objective is to examine whether the three poets exhibit measurable differences in phonological density, as reflected by the average number of syllables and primary stresses per line. Through these quantitative phonological measures, the analysis provides an initial comparison of rhythmic characteristics across the examined corpora.

From a methodological perspective, the study draws upon computational phonological analysis, an approach concerned with the quantitative examination of the sound structure of language through measurable phonetic and prosodic features. In contrast to traditional literary approaches, computational methods enable the systematic identification and comparison of sound-based patterns across texts.

Within this framework, particular attention is given to *phonological density*, which refers to the concentration of sound patterns within a poetic text. As an indicator of auditory complexity, it contributes to the rhythm and musicality of poetry. Accordingly, the present analysis explores syllabic and stress-based variation in order to identify stylistic differences between the three poets.

II. DATA MANAGEMENT

A. Corpus Overview

The corpus employed in the present study consists of poetic texts authored by Shakespeare, Dickinson, and Browning. Provided by the course instructor, the texts serve as representative examples of each poet's work and form a small corpus suitable for the exploratory examination of phonological characteristics.

B. Data Preprocessing

Prior to analysis, a preprocessing pipeline was implemented to facilitate phonological feature extraction. Specifically, all text was converted to lowercase, punctuation was removed, and each line was tokenised into individual words. Phonological information was subsequently obtained from the *Carnegie Mellon University (CMU) Pronouncing Dictionary*, which maps English words to their phonetic transcriptions and corresponding stress patterns. These phonetic representations were then used to derive syllable and primary stress counts for each poetic line, forming the basis of the subsequent phonological analysis.

III. METHODOLOGY

A. Phonological Feature Extraction

The analysis was based on two complementary measures: **Average Syllables per Line (ASL)** and **Average Stresses per Line (ASL)**, which jointly capture aspects of phonological density and rhythmic structure.

1) *Average Syllables per Line (ASL)*: ASL measures the average number of syllables contained in a poetic line. As an indicator of phonological density, it reflects the extent to which a poet's language is compact or extended, while also influencing rhythm, flow and breath during poetic delivery.

$$ASL = \frac{\sum \text{Syllables}}{N_{\text{lines}}} \quad (1)$$

where $\sum \text{Syllables}$ denotes the total number of syllables and N_{lines} the total number of poetic lines.

2) *Average Stresses per Line (AStL)*: AStL measures the average number of primary stresses occurring within a poetic line. As a fundamental component of poetic meter, it provides an indication of the strength and regularity of rhythmic structure within a poem.

$$AStL = \frac{\sum Stresses}{N_{lines}} \quad (2)$$

where $\sum Stresses$ denotes the total number of primary stresses and N_{lines} the total number of poetic lines.

B. Statistical Analysis and Visualisation

For both phonological features, descriptive statistics were calculated to provide an overview of the central tendency and dispersion of the distributions. In addition, boxplots and histograms were employed to visualise the distributions and highlight potential differences in phonological characteristics across the examined corpora.

IV. RESULTS AND FINDINGS

The phonological characteristics of the poetic corpus were examined through syllable and stress distributions. The descriptive statistics and visualisations provide an overview of the similarities and differences observed across the examined poets (Figures 1 and 2).

Shakespeare exhibits relatively high syllable counts and comparatively elevated stress counts per line, with both distributions appearing moderately concentrated around their central tendencies. The visualisations indicate a generally consistent phonological profile, accompanied by a noticeable number of extreme observations.

Dickinson is characterised by lower syllable and stress counts overall. In addition, her syllable distribution appears more dispersed than those of the other poets, reflecting greater variability across poetic lines.

Browning presents the highest overall syllable and stress counts, while her distributions remain relatively concentrated. The graphical representations suggest a consistent phonological profile with comparatively limited dispersion.

Overall, Shakespeare and Browning display similar phonological patterns, characterised by higher syllable and stress counts, whereas Dickinson exhibits consistently

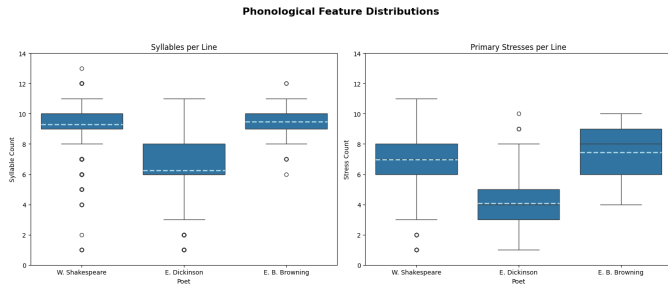


Fig. 1: Phonological feature distributions across poets.

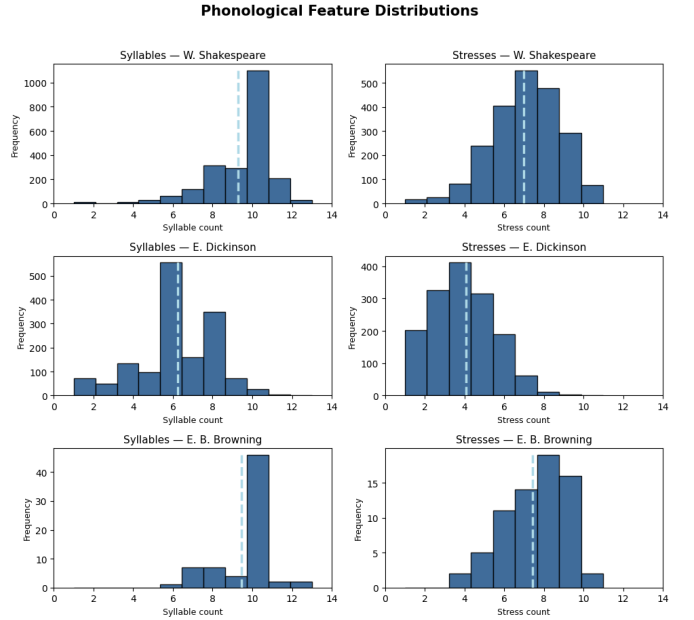


Fig. 2: Phonological feature distributions across poets.

lower values. visualisations further reveal differences in variability, with Dickinson showing greater dispersion, particularly with respect to syllable counts, and Browning exhibiting the most concentrated distributions among the three corpora.

V. INTERPRETATION AND DISCUSSION

A. Which Poet Exhibits Higher Phonological Density?

The findings suggest that Browning exhibits the highest phonological density, closely followed by Shakespeare. Both poets display higher syllable and stress counts per line, indicating a denser and more metrically structured poetic style. In contrast, Dickinson presents a more compact and minimalist form of expression, characterised by shorter poetic lines and lower stress frequencies.

B. Which Poet Exhibits Greater Variability?

Dickinson exhibits the greatest variability, particularly with respect to syllable counts, indicating a less uniform phonological structure across lines. By contrast, Browning displays the most consistent distributions, while Shakespeare occupies an intermediate position. Overall, Browning's poetry appears more regular and structured, whereas Dickinson's verse demonstrates greater variation in its phonological organisation.

VI. CONCLUSION

Overall, the analysis demonstrates the value of computational phonological measures for the quantitative comparison of poetic styles, revealing meaningful differences in rhythmic structure and phonological density. Nevertheless, the present study should be viewed as exploratory in nature. Consequently, the analysis was intentionally

limited to syllabic and stress-based measures, while additional dimensions of phonological density, such as rhyme frequency, were not examined and may be incorporated in future investigations to provide a more comprehensive representation of poetic sound patterns.